

Kyle Marino

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EDUCATION

University of Southern California ■ Los Angeles, CA

2019 - 2024

- **M.S. Computer Engineering 2024 (4.0)**
- **B.S. Electrical and Computer Engineering 2023 (3.93)** - Presidential Scholarship, Summa Cum Laude
- Relevant Coursework: Computational Intelligence and Neural Learning, Deep Learning, Internet and Cloud Computing, Computer Systems Organization, SOC Design, Probability Theory, Linear Algebra

WORK / RESEARCH EXPERIENCE

FPGA Architecture Intern ■ Apple

May 2024 – Sep 2024

- Developed Python driver and Verilog AXI interface for SPI DAC peripheral on MIX FPGA.
- Integrated DisplayPort IP at 8k 60 Hz over CIO80 on Agilex7 FPGA with hardware testing platform.
- Wrote a SERDES link training module and PRBS generator in Verilog for loopback testing at 1.5 GHz.

Directed Research Under Dr. Viktor K. Prasanna ■ USC

Feb 2023 – May 2024

- Designed a Vision Transformer accelerator for FPGAs, achieving up to a 17.89x reduction in memory bandwidth and a 2.16x improvement in throughput per DSP over SOTA designs.
- **Kyle Marino**, Pengmiao Zhang, & Viktor K. Prasanna. ME-ViT: A Single-Load Memory-Efficient FPGA Accelerator for Vision Transformers. *High Performance Computing (HiPC)*, 2023. **Best Paper Award**

RTL Integration Intern ■ AMD/Xilinx

May 2023 – Aug 2023

- Built a distributed design rule verification system (1000+ jobs) in Python to analyze full-chip RTL.

SOC Integration Intern ■ AMD/Xilinx

May 2022 – Aug 2022

- Designed and integrated Python and Perl tools for automated EMIR reporting and simulation processes.

Computational Intelligence and Neural Learning Final Project ■ EE 689

Aug 2023 – Dec 2023

- Developed a novel method for generating adversarial attacks to force specific language model token outputs for Llama 2 7B using Python and CUDA.

Avionics Software and FPGA Engineer ■ USC Rocket Propulsion Lab

Sep 2019 – Jan 2023

- Built a 13 channel data collection and processing platform on Intel MAX10 FPGA recording at 5 MB/s.
- Designed a custom microprocessor and Python compiler for real-time quaternion integration of IMU data.
- Wrote C++ drivers to interface with FPGA and battery management boards.

Internet and Cloud Computing Class Projects ■ EE 542

Aug 2022 - Dec 2022

- Designed a custom file transport protocol with multithreaded UDP sockets that outperforms TCP over a high loss link, achieving over 75 Mbps out of 80 Mbps theoretical.
- Modified TCP Linux kernel to improve throughput over a high loss link, from 480 Kbps to 10 Mbps.

LEADERSHIP

Captain ■ USC Climbing Team

Aug 2021 – May 2024

- Lead practices for competition training, mentored members on technique, strength training, and injury rehab.

PERSONAL PROJECTS

Climbing Training App

Ian 2024 – Current

- Swift app for tracking and planning workouts centered around injury rehabilitation.

3D Physics Engine

Nov 2019 - Dec 2019

- A GPU-accelerated, real-time physics and rendering engine in Java with LWJGL and developed a custom collision-detection algorithm for any arbitrary object model.

Quadcopter

Dec 2016 – Jun 2019

- Self-stabilizing quadcopter with video streaming capabilities, with web interface for remote control.

OTHER

- Programming Skills: Python, LLMs, Vision Transformers, C/C++, Verilog, Java, Javascript, SQL, Mongo DB
- Activities: Eagle Scout, Rock Climbing, Guitar, Kendama